MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE AND FIRE RESISTANT, LIGHT WEIGHT, SMALL DIAMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the wire described herein shall consist of this specification and the latest issue of MIL-W-25038.

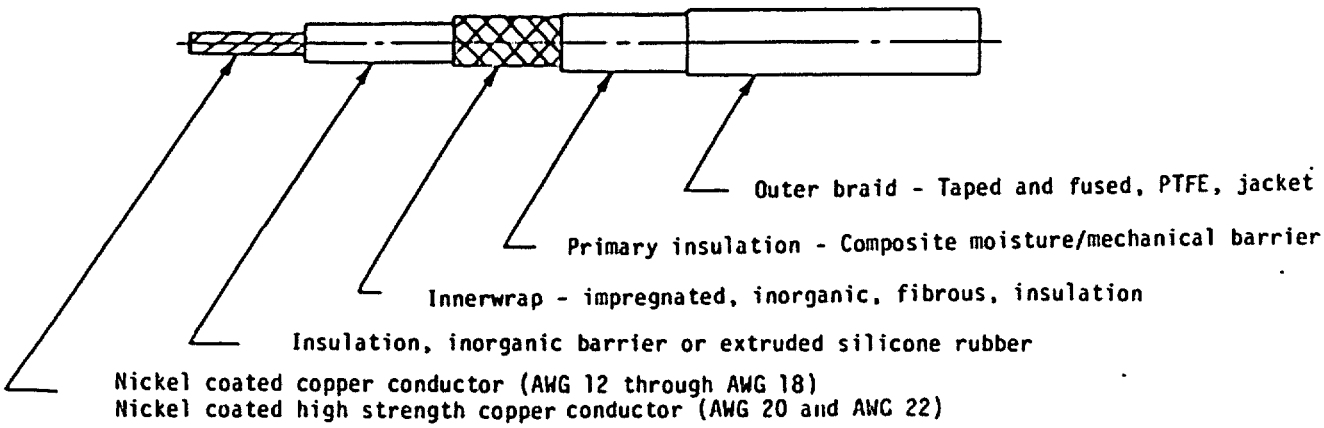


FIGURE 1. Design and construction.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 V rms.

Thermal shock: 0.60 inch maximum change in insulation measurement.

Durability of identification: 250 cycles (500 strokes).

Color: Unless otherwise specified by the part number, the color shall be white.

Part number: M25038/3- (wire size from table I)-(color code designator in accordance with MIL-STD-681).

Examples:
Size 20 AWG, white = M25039/3-20-9.
Size 20 AWG, white with orange stripe = M25038/3-20-93.

Note: Materials used in construction of the wire must be identified and approved by the qualifying activity. Wire construction cannot contain asbestos.

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TABLE I. Finished wire construction and performance details.

Wire size 1/	Conductor strand 2/			Maximum conductor diameter (inch)	Maximum dc resistance 1,000 ft. at 20°C (ohms)	Finished wire diameter (max) (inch) 3/	Weight 1,000 ft. (pounds) (max)	Performance details	
	Number of strands	AWG size	Nominal diameter (inch)					Bend test mandrel OD (inches)	Bend test mandrel Load (pounds)
22	19	34	.0063	.033	23.70	.054	4.15	6	2
20	19	32	.008	.041	15.27	.083	9.0	6	2
18	19	30	.010	.052	8.5	.097	10.5	6	2
16	19	29	.0113	.061	6.66	.103	13.5	8	3
14	19	27	.0142	.074	4.32	.123	19.5	8	3
12	19	25	.0179	.093	2.78	.142	28.0	8	3

1/ Wire size 20 AWG and 22 AWG shall be constructed with nickel coated high strength copper strands. Conductor tensil strength and elongation test methods and requirements for these sizes are to be determined.

2/ The conductor strands shall have an annulor nickel coating, which its weight shall not be less than 27 percent of the total weight of the nickel-coated strand when tested in accordance with ASTM B355.

3/ These wires comply with the sealing range requirements for firewall connectors as specified in MIL-C-5015, MIL-C-38999, and MIL-C-83723.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES
NASA
IS

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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